



<https://media.sweetwater.com/m/insync/2018/08/keyboardexpression.jpg>

Adapter For Expression Pedals

Amon Benson | Sensor and Actuator Systems | Milestone Meeting 1 | 27. September 2026

Motivation

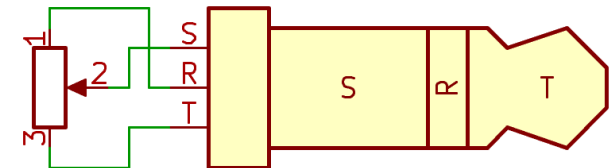
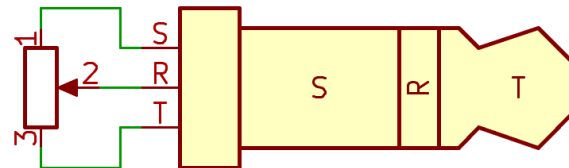
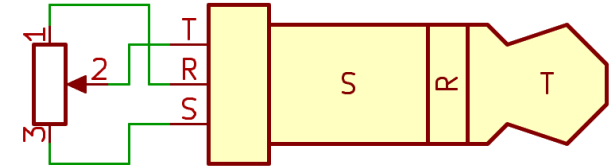
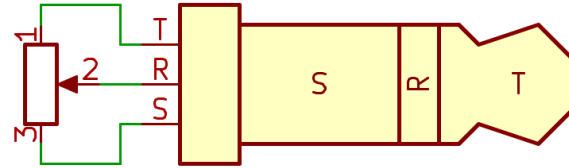


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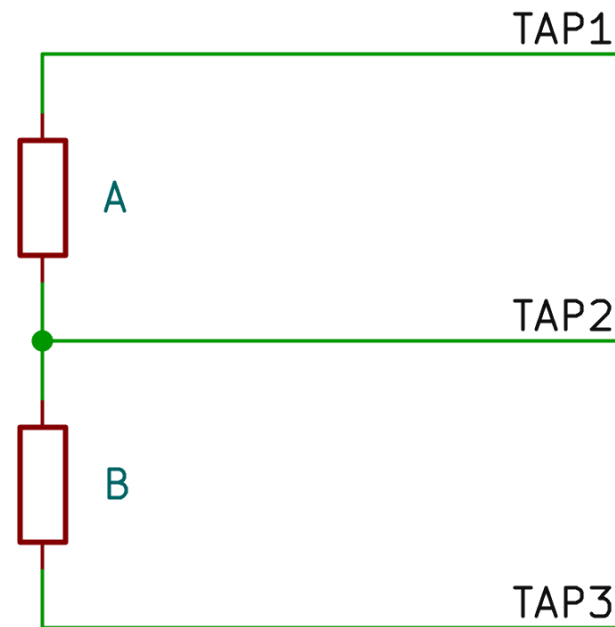
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Motivation

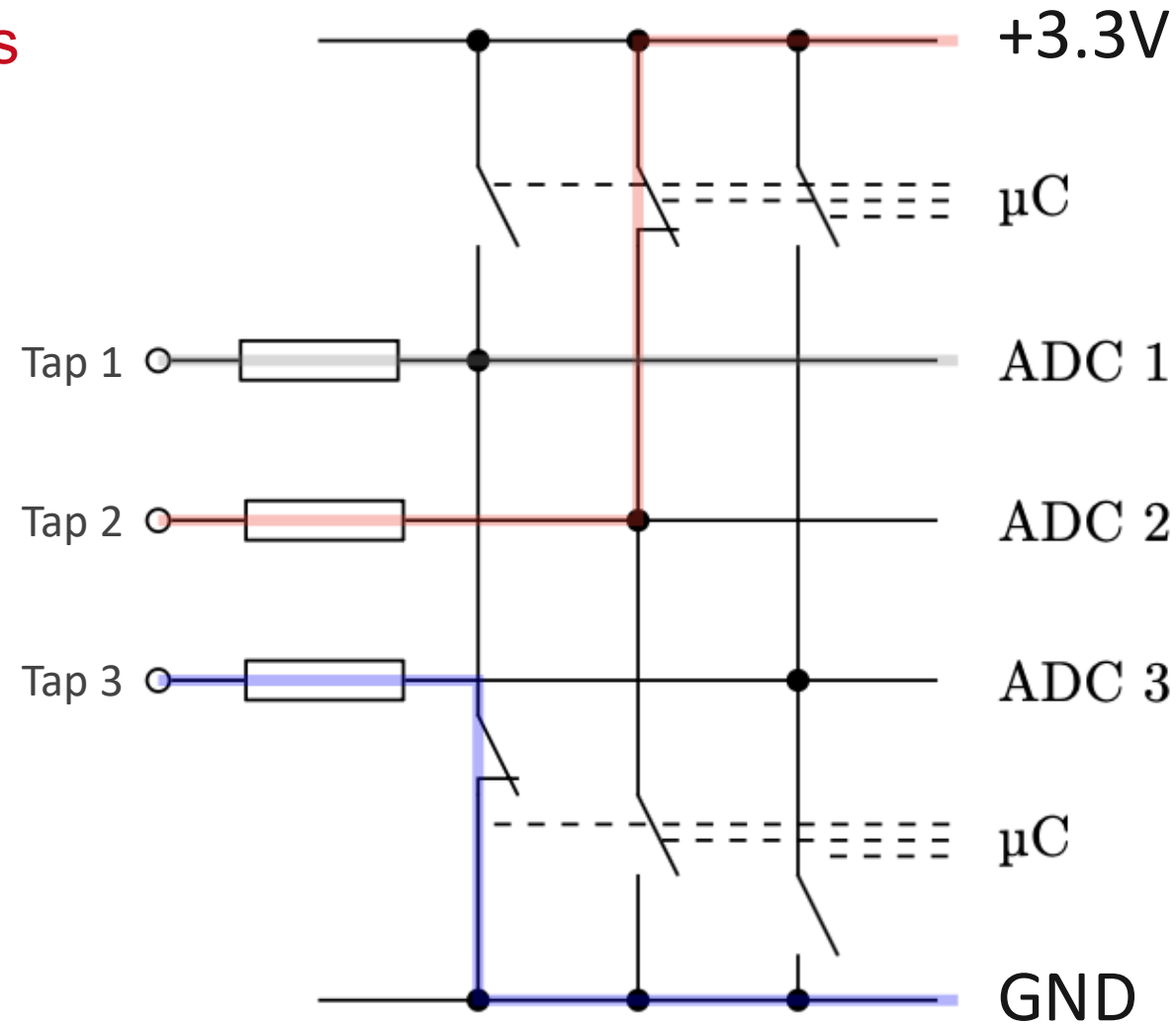


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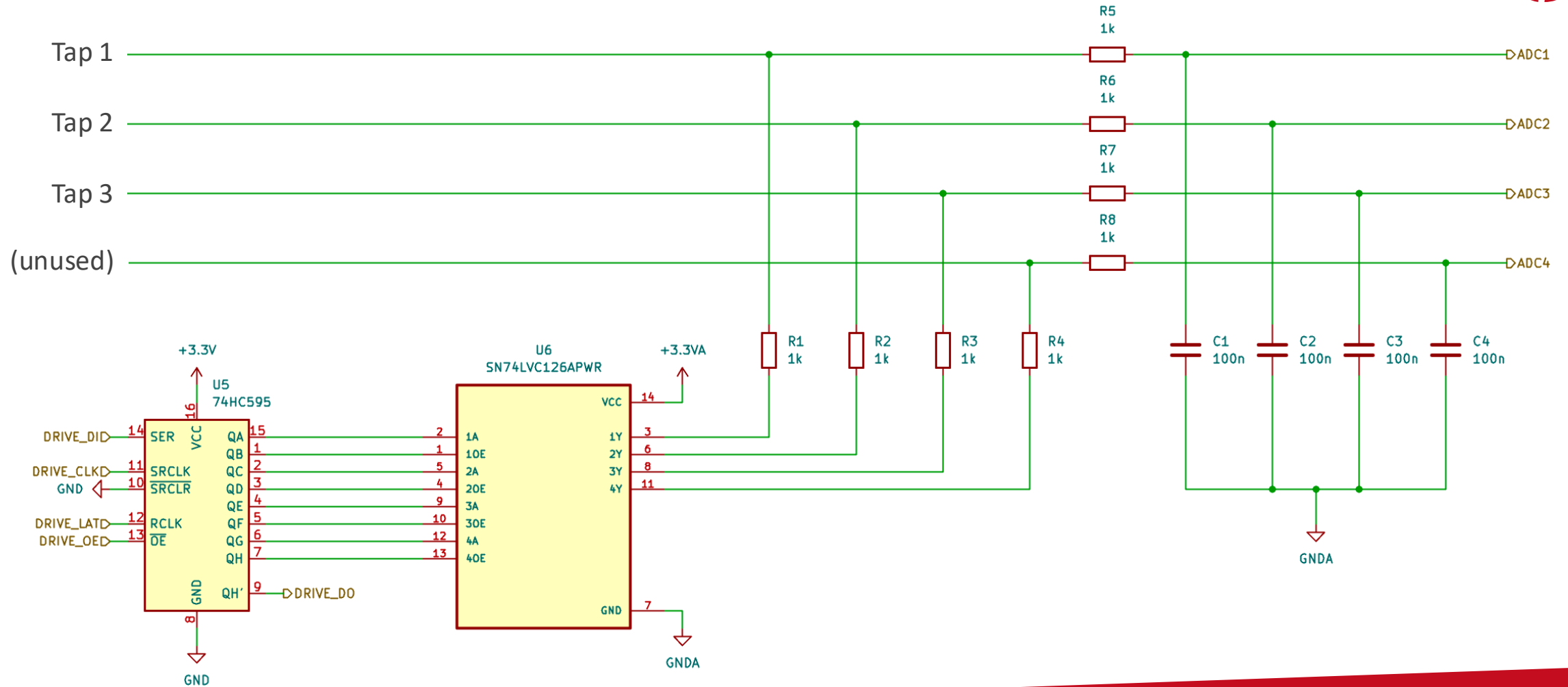
Detection Process



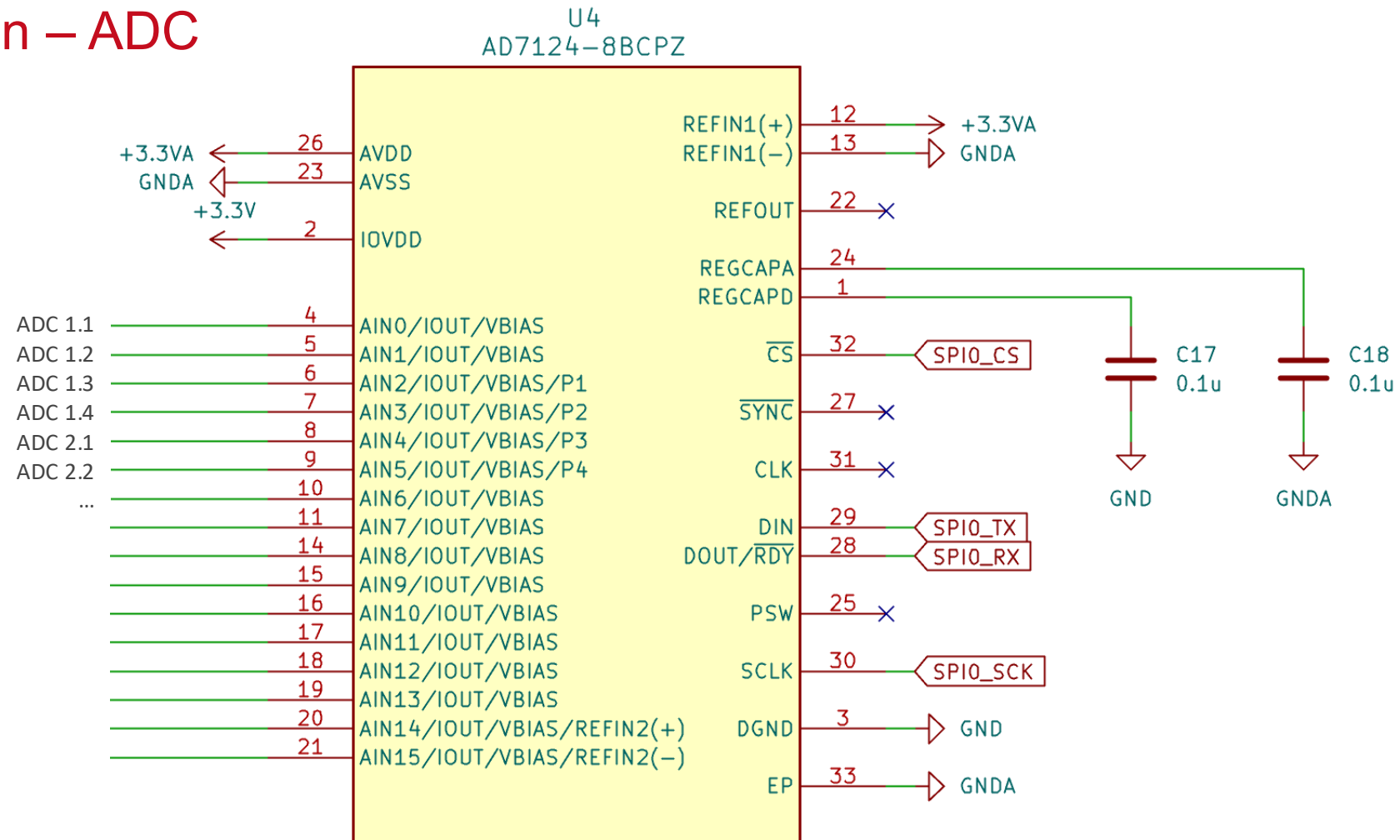
Detection Process



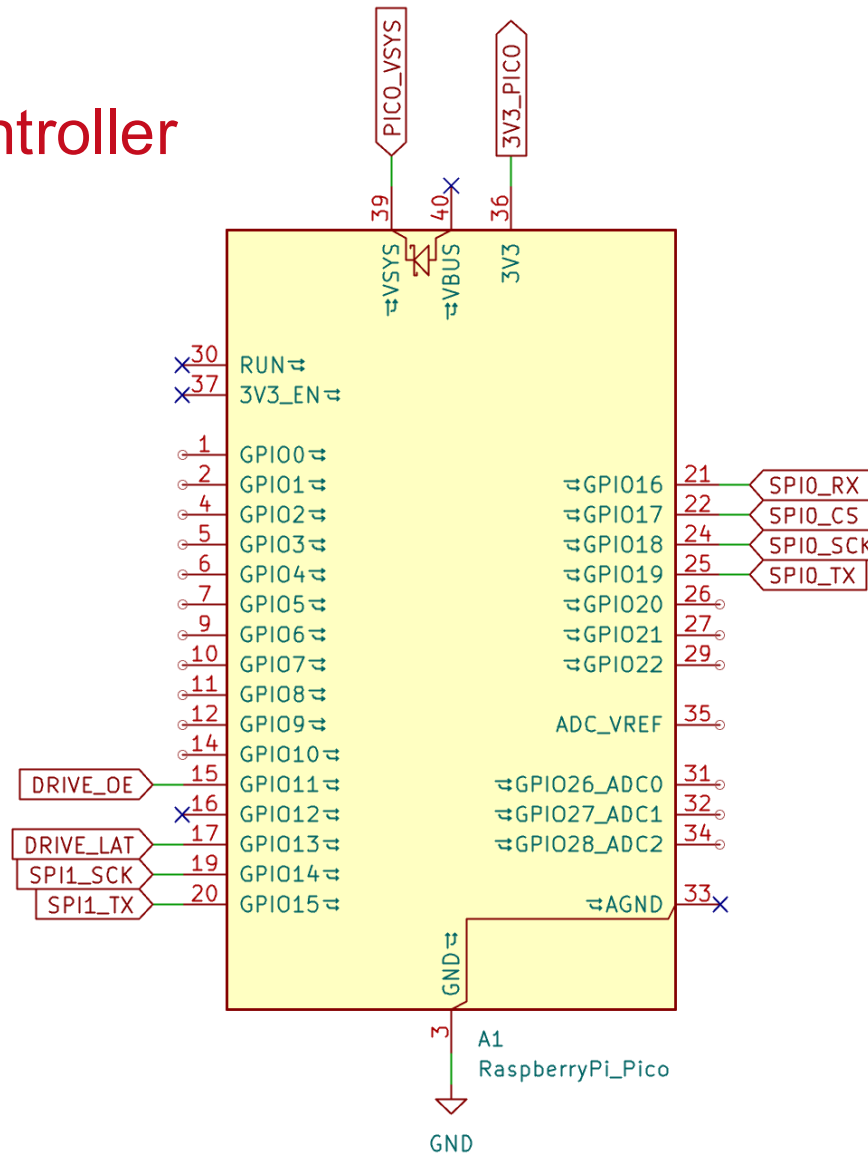
Circuit Design – Analog Frontend



Circuit Design – ADC

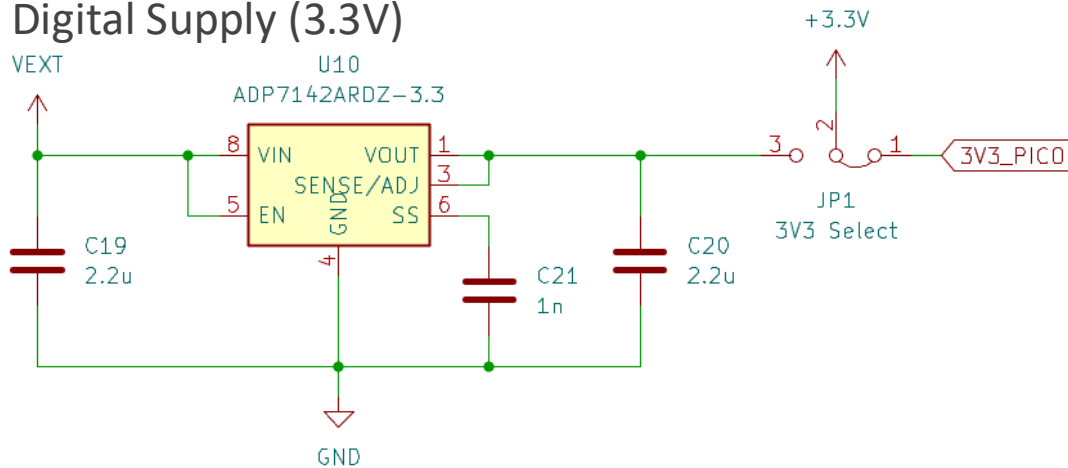


Circuit Design – Microcontroller

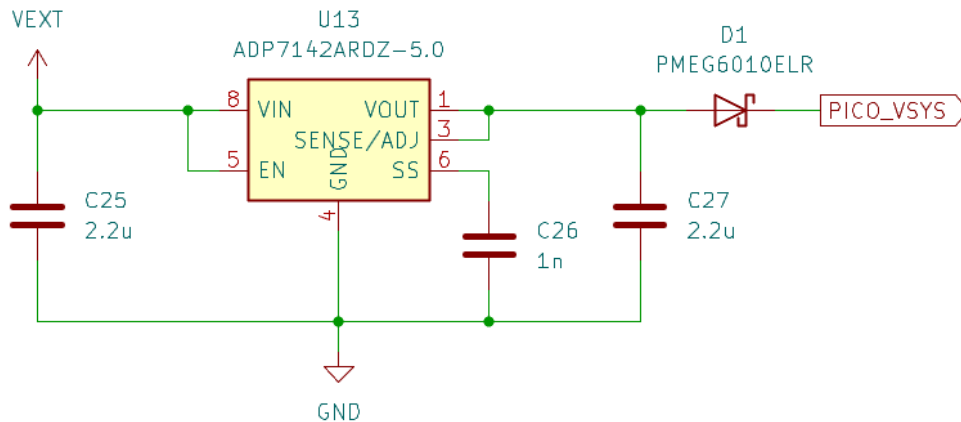


Circuit Design – Power Supply

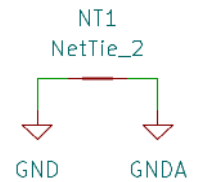
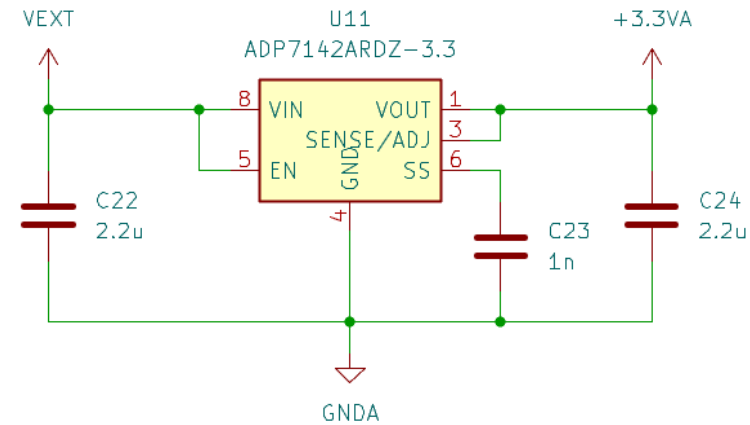
Digital Supply (3.3V)



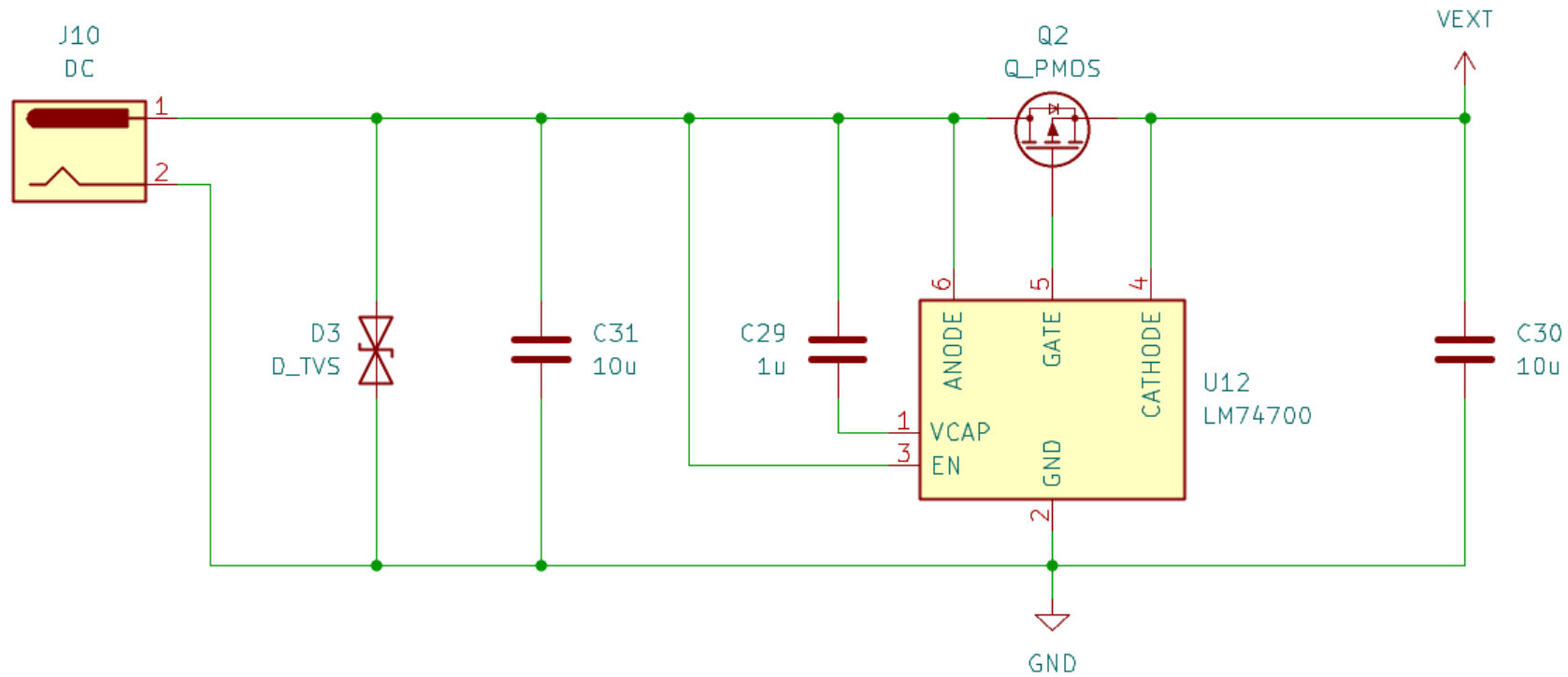
Digital Supply (5V)



Analog Supply (3.3V)



Circuit Design – DC Input Protection



Outlook

- Hardware
 - ~~— circuit design~~
 - PCB layout and routing
 - manufacturing
- Firmware
 - ~~— planning of the detection algorithm~~
 - implementation using the microcontroller toolchain
 - simulation of ADC measurements for offline (functional) testing

Testing and Evaluation Plan

- power supply quality
- SPI signal quality
- functional self-test
 - SPI device inquiry (ADC)
 - tristate buffer
- detection algorithm for known static networks
- accuracy and linearity
- end-to-end latency

Sources

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- [2] Mission Engineering, "Understanding expression pedals." [Online]. Available: <https://missionengineering.com/understanding-expression-pedals/>
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“AVR-Transistortester”



<https://content.instructables.com/F4V/9O1Y/IFCDEPNY/F4V9O1YIFCDEPNY.jpg>